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(54) **ONE-PIECE CAP, SPOUT, AND AIR VENT**
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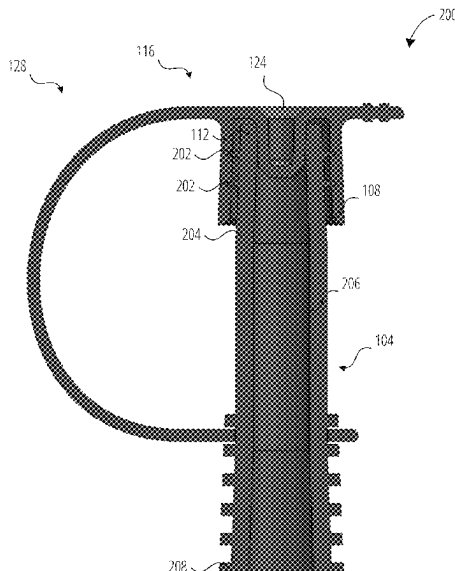
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(57) **ABSTRACT**

Disclosed herein is a one-piece spout device and a method of making the same. The one-piece spout device has a spout body, cap portion, and an air-vent portion. A method of making a one-piece spout device involves heating a resilient flexible material and molding it into the shape of a one-piece spout device. The one-piece spout device preferably provides an easy-to-use spout attachable to containers, including gas cans, which does not leak and remains resilient and flexible over a wide range of temperatures, and is long-lasting with repeated use.

11 Claims, 5 Drawing Sheets



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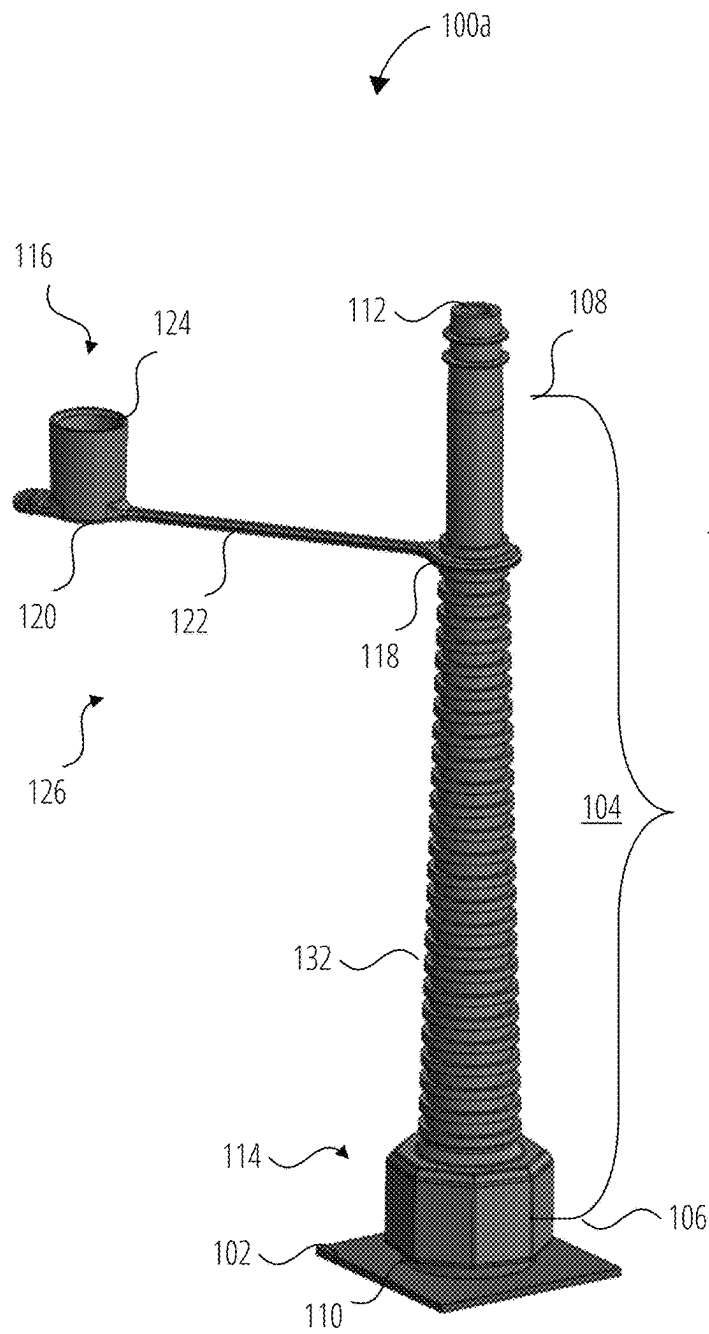


FIG. 1A

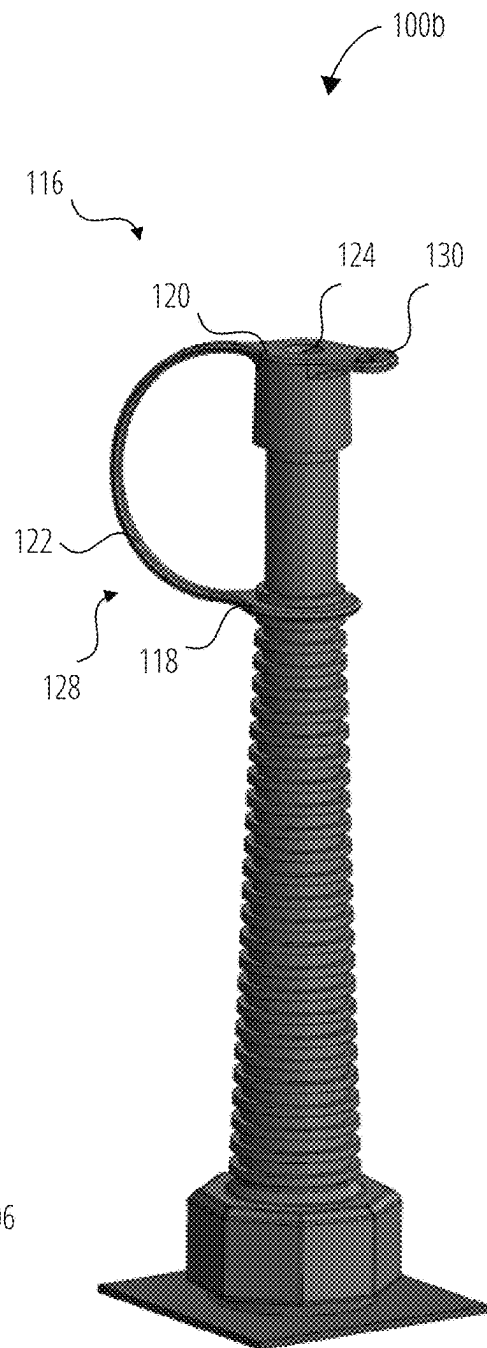


FIG. 1B

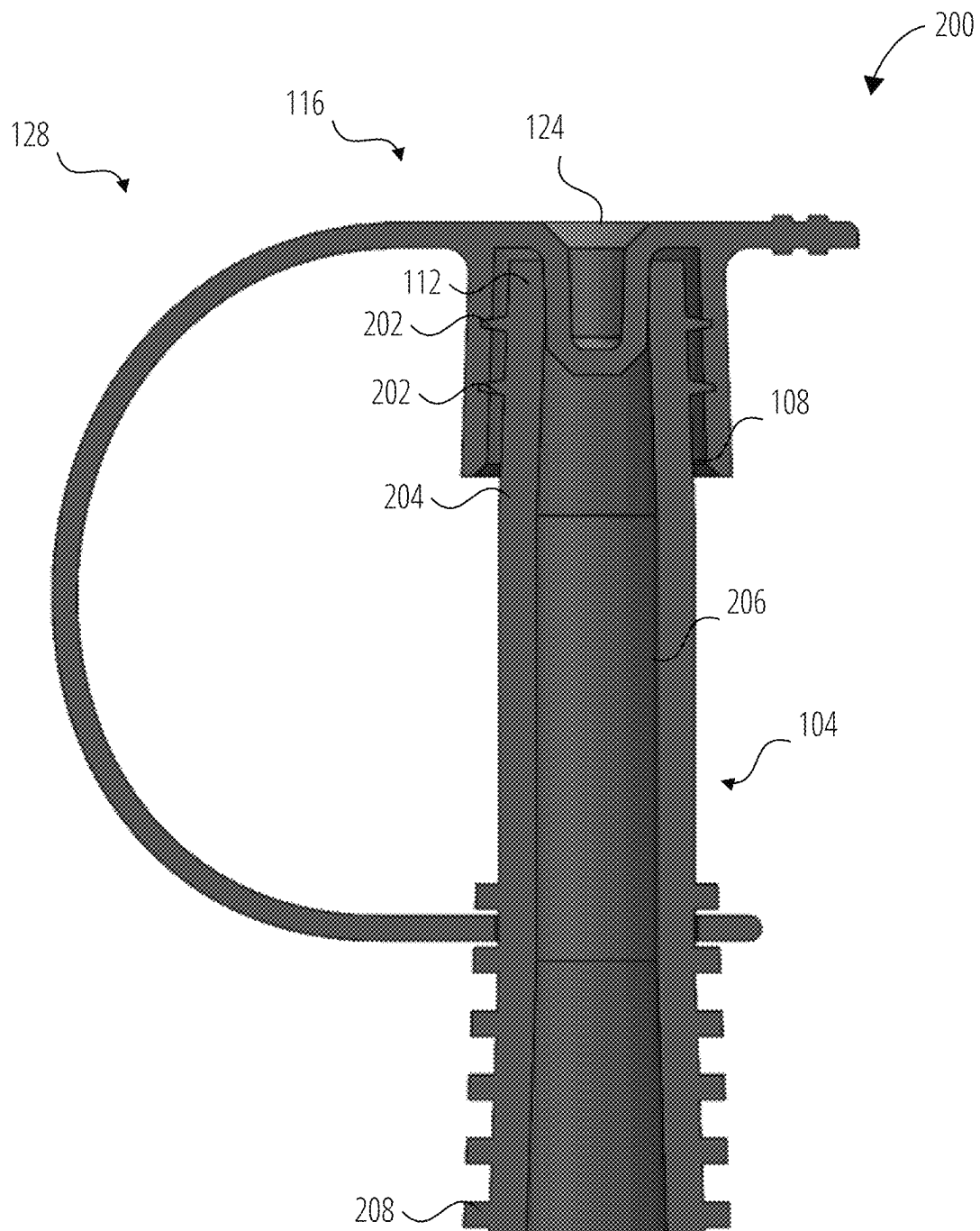


FIG. 2

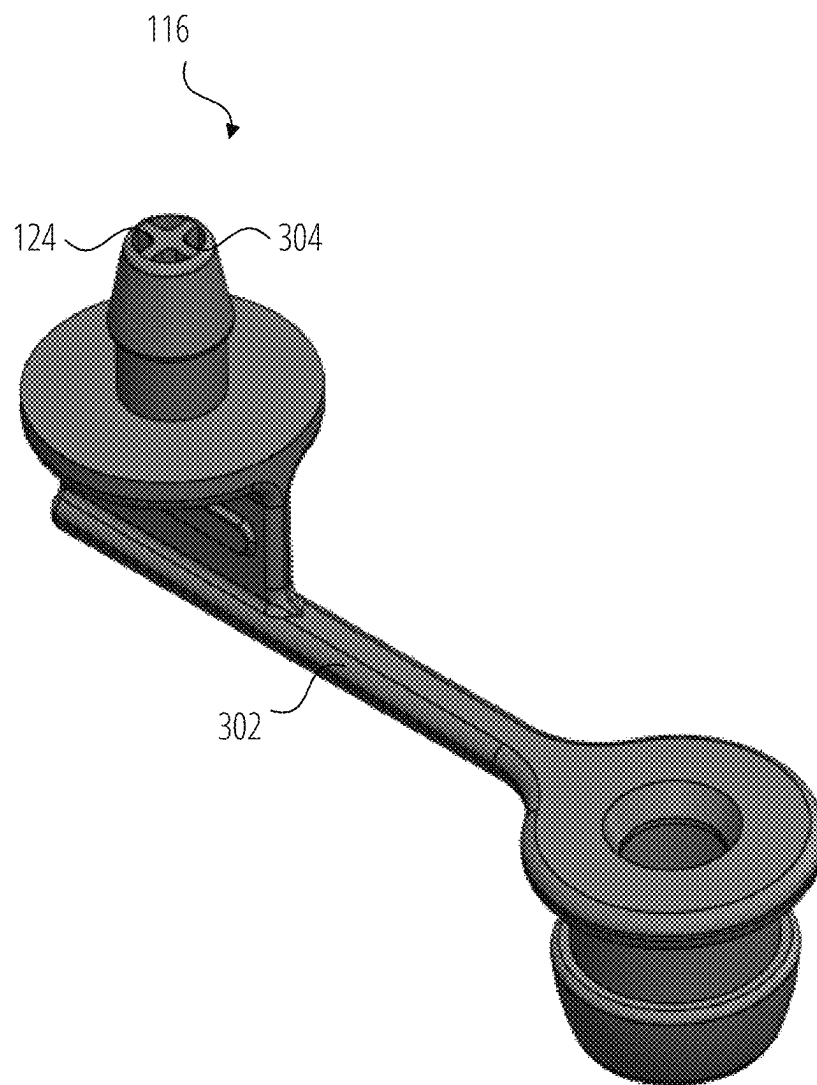


FIG. 3

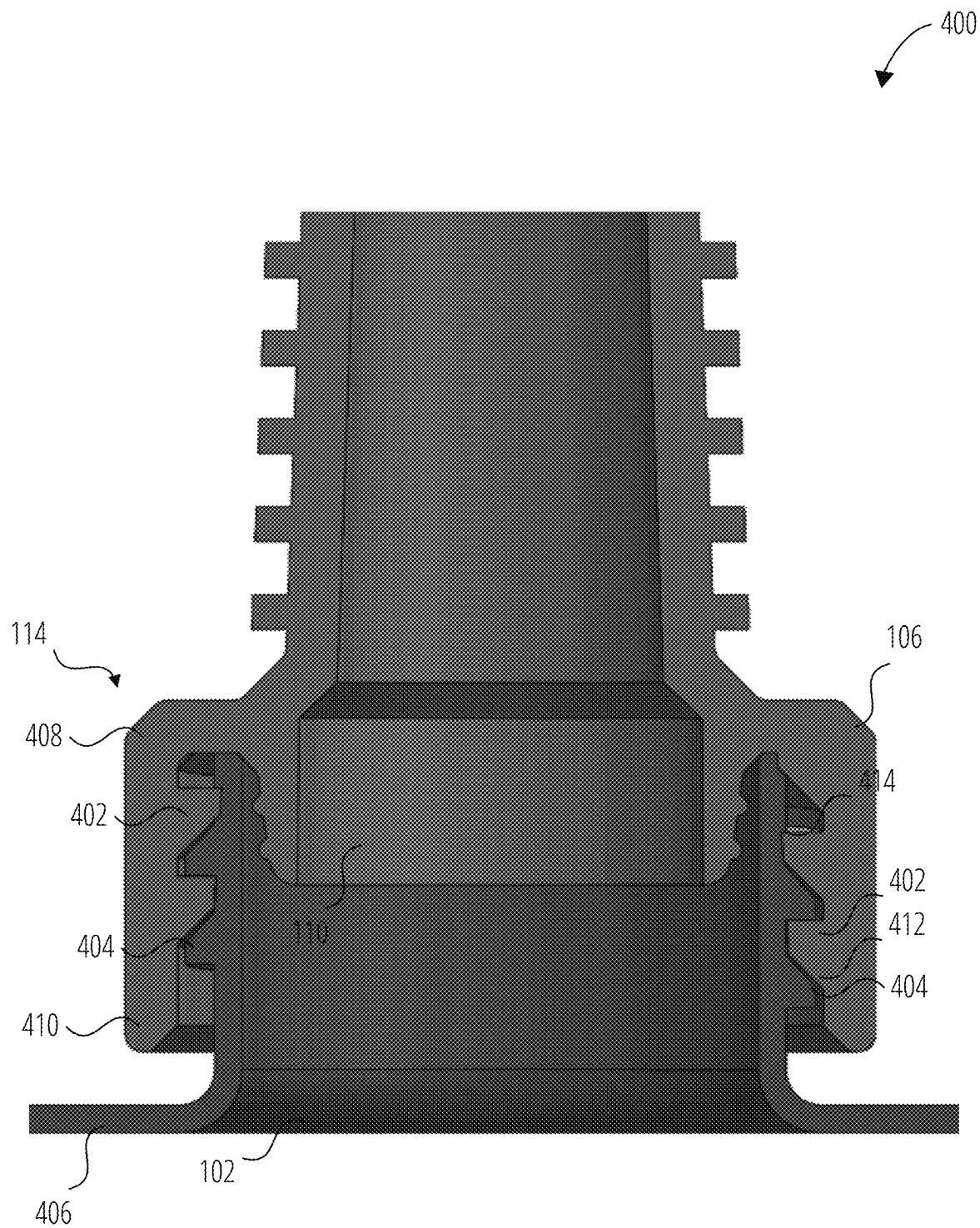


FIG. 4

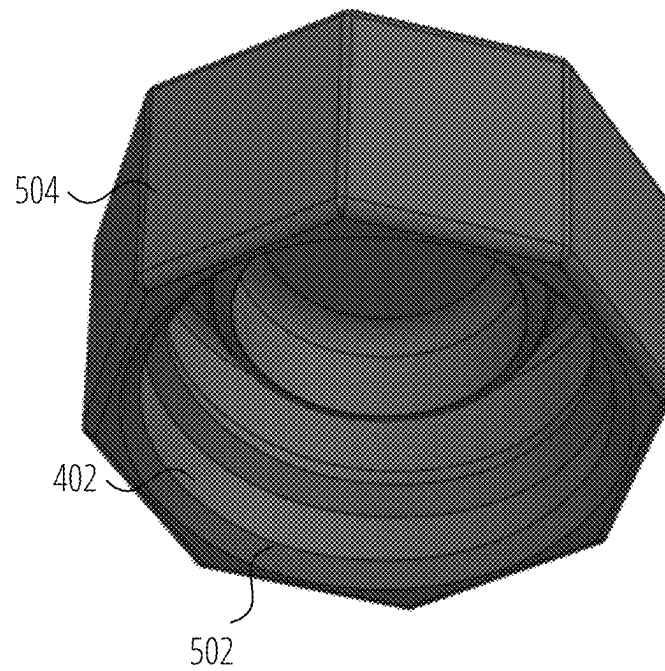


FIG. 5

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ONE-PIECE CAP, SPOUT, AND AIR VENT**FIELD OF THE INVENTION**

The present invention relates generally to the field of automobiles, hardware, and coverings for containers, and more particularly for lids, caps, spouts, and air vents for containers, and more specifically to one-piece spouts, caps, and air vents for fluid and gas cans.

BACKGROUND

In the prior art, a plastic spouts for fluid containers and gas cans may be obtained at various retailers in an area; such as outlets, big box stores, hardware and retailers.

Prior art and fluid containers gas cans generally comprise old fashioned plastic spouts, which typically comprise multiple pieces, including, but not limited to a spout, gasket, a cap.

These multi-piece plastic spouts can be difficult to assemble, tend to get brittle in cold weather or a cold environment, and they can leak when assembled, which can waste fluids and could pollute the environment.

Typically, in the prior art, plastic gas cans or fluid containers have been provided without an air vent.

The multi-piece plastic spouts can result in lost pieces.

The multi-piece plastic spouts generally tend to be fragile and breakable, and wear out over time with repeated use.

In the prior art, spouts may not have been configured to adapt to receptacles of older vehicles.

Many gas can users, such as people or contractors, with small gasoline engines for their cars, older automobiles, lawnmower, and the like, have experienced the aforementioned deficiencies faults and problems with current multi-piece fluid spouts.

Hence, there is a need in the art for a spout which overcomes one or more of the above deficiencies.

There is a need for a one-piece, spout, cap, and air vent for fluid containers, including gas cans, which requires no assembly, is flexible and adaptable, seals tightly to a container without the need for gaskets or o-rings, and does not leak.

BRIEF SUMMARY

It is an object of the invention to provide a one-piece cap, spout, and air vent.

According to an aspect of the invention, there is provided a one-piece spout device, attachable to a container opening, the one-piece spout comprising: a spout body having a bottom end with a bottom opening and a top end with a top opening; cap portion affixed to the bottom end, the cap portion being sized and adapted to releasably connect said one-piece spout to said container; and an air-vent portion comprising a first end and a second end, the first end being affixed to the top end of said spout body, and the second end being attachable to the top opening of said top opening, the second end comprising a vent opening for partially covering said top opening; wherein the first end and the second end of said air-vent portion are attached by a tether element.

According to an aspect of the invention, there is provided a method of preparing a one-piece spout from a mold, the method comprising: heating a resilient flexible material; pouring said heated resilient flexible material into a negative mold of a one-piece spout, the mold comprising a negative indent of: a spout body having a bottom end with a bottom opening and a top end with a top opening; a cap portion

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affixed to the bottom end, the cap portion being sized and adapted to releasably connect said one-piece spout to said container; an air-vent portion comprising a first end and a second end, the first end being affixed to the top end of said spout body, and the second end being attachable to the top opening of said top opening, the second end comprising a vent opening for partially covering said top opening, wherein the first end and the second end of said air-vent portion are attached by a tether element; and, allowing said heated resilient flexible material to cool to room temperature.

The advantages and features of the present invention will become better understood with reference to the following more detailed description and claims taken in conjunction with the accompanying drawings in which like elements are identified with like symbols.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

In the figures, embodiments are illustrated by way of example. It is to be expressly understood that the description and figures are only for the purpose of illustration and as an aid to understanding.

Embodiments will now be described, by way of example only, with reference to the attached figures, wherein the figures:

FIG. 1A illustrates an aspect of the subject matter in accordance with one embodiment.

FIG. 1B illustrates an aspect of the subject matter in accordance with one embodiment.

FIG. 2 illustrates an aspect of the subject matter in accordance with one embodiment.

FIG. 3 illustrates an aspect of the subject matter in accordance with one embodiment.

FIG. 4 illustrates an aspect of the subject matter in accordance with one embodiment.

FIG. 5 illustrates an aspect of the subject matter in accordance with one embodiment.

DETAILED DESCRIPTION

The term “connected”, “attached”, “affixed” or “coupled to” may include both direct coupling (in which two elements that are coupled to each other contact each other) and indirect coupling (in which at least one additional element is located between the two elements).

Devices and methods for carrying out the invention are presented in terms of embodiments depicted within the FIGS. However, the invention is not limited to the described embodiments, and a person skilled in the art will appreciate that many other embodiments of the invention are possible without deviating from the basic concept of the invention, and that any such work around will also fall under scope of this invention. It is envisioned that other styles and configurations of the present invention can be easily incorporated into the teachings of the present invention, and the configurations shall be shown and described for purposes of clarity and disclosure and not by way of limitation of scope.

The features of the invention which are believed to be novel are particularly pointed out in the specification. The present invention now will be described more fully hereinafter with reference to the accompanying drawings, which

are intended to be read in conjunction with both this summary, the detailed description and any preferred and/or particular embodiments specifically discussed or otherwise disclosed. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided by way of illustration only and so that this disclosure will be thorough, complete and will fully convey the full scope of the invention to those skilled in the art.

FIG. 1A illustrates an embodiment of a one-piece spout device **100a**, attachable to a container opening **102**, the one-piece spout device **100a** preferably comprising: a spout body **104** having a bottom end **106** with a bottom opening **110** and a top end **108** with a top opening **112**; a cap portion **114** affixed to the bottom end **106**, the cap portion **114** being sized and adapted to releasably connect said one-piece spout device **100a** to said container opening **102**.

The one piece one-piece spout device **100a** preferably further comprises an air-vent portion **116** including a first end **118** and a second end **120**, the first end **118** being affixed to the top end **108** of said spout body **104**, and the second end **120** being attachable to said top opening **112**, the second end **120** comprising a vent opening **124** for partially covering said top opening **112**.

Preferably, the first end **118** and the second end **120** of said air-vent portion **116** are attached by a tether element **122**.

Preferably, the top opening **112** is configured for insertion into a vessel, for example a gasoline engine or water tank.

Preferably, the top opening **112** and the spout body **104** are sized and configured to fit into a variety of different vessels, including, but not limited to, older gasoline engines in cars, newer gasoline engines in cars, small gasoline engines (lawn mowers, for example), and the like.

In an open configuration **126**, the top opening **112** of the one-piece spout device **100a** may be uncovered by the air-vent portion **116**. When uncovered, fluid may flow from a container opening **102**, through the spout body **104**, and in to a vessel.

The one-piece spout device is releasably attachable at a cap portion **114** to a container opening **102**, wherein the container opening **102** is preferably an opening on a container, tank, vessel, and the like.

The cap portion **114** releasably attaches to the container opening **102** by twisting around the container opening **102**. This can be seen in greater detail in FIG. 4.

Preferably, the one-piece spout device comprises a resilient flexible material **132**. Preferably, each component of the one-piece spout device is made from the same resilient flexible material **132**, the resilient flexible material **132** enabling the one-piece spout device **200** to remain flexible and stable over a wide range of temperatures.

A resilient flexible material **132** may include, but is not limited to flexible plastics, polymers, and rubbers.

Preferably, the resilient flexible material **132** is resistant to oils and acids, and is strong and flexible.

Rubbers may include nitrile rubber, a synthetic rubber derived from acrylonitrile (ACN) and butadiene. Nitrile rubber may also known as nitrile butadiene rubber, NBR, Buna-N, and acrylonitrile butadiene rubber.

Preferably, a wide range of temperatures includes temperatures above and below freezing. For example, nitrile rubber remains flexible and stable over a temperature range of -40 to 108°C ., and is generally resistant to oil, fuel, and

other chemicals. Nitrile rubber is more resistant than natural rubber to oils and acids, and has superior strength, but has inferior flexibility.

Furthermore, nitrile rubber used in this capacity is novel in that one-piece spout devices **200** formed from nitrile rubber may be can be compatible with other devices in the automotive and aeronautical industry; including, but not limited to, fuel and oil handling hoses, seals, grommets, and self-sealing fuel tanks, molded goods, footwear, adhesives, sealants, sponges, expanded foams, and floor mats.

Furthermore, one-piece spout devices **200** formed from nitrile rubber may be can be compatible with other devices used in the nuclear industry. It is used in the nuclear industry to make protective gloves. NBR's stability at temperatures from -40 to 108°C . (-40 to 226°F .) makes it an ideal material for aeronautical applications.

Nitrile butadiene is also used for disposable lab, cleaning, and examination gloves.

The one-piece spout devices **100a** is preferably of a one-piece molded and solid construction.

The spout body **104** is preferably a conical shape, adaptable for insertion in a vessel or tank, for filling the vessel or tank with fluids, which preferably flow from a container, through the one-piece spout device **100a**, and into the vessel.

When the one-piece spout device **100a** is releasably attached to a container, fluid generally flows from the container, into the one-piece spout device **100a** at the cap portion **114**, through the bottom opening **110**, along the spout body **104**, through the top opening **112**, and into a vessel or simply out (onto the ground, or into an open container, for example).

The one-piece spout device is preferably releasably attached to a container/tank for pouring fluids from the container/tank without leaking any fluids.

A method of preparing the one-piece spout devices **100a** from a mold preferably comprises heating a resilient flexible material **132**, pouring the material into a negative mold of the one-piece spout device **100a**, the mold comprising a negative indent of the one-piece spout device, and allowing said heated nitrile rubber to cool to room temperature.

Preferably, but not necessarily, the mold is a solid metal (steel, for example) mold.

Alternatively, the method may be performed with a positive mold of the one-piece spout device **100a**, if desired.

FIG. 1B illustrates a one-piece spout device **100a** wherein the top opening **112** is partially covered with the second end **120** of said air-vent portion **116**.

In a closed configuration **128**, the second end **120** of the air-vent portion **116** engages with the top opening **112** by being pushed into the top opening **112**. Air may enter through a vent opening **124** on the second end **120**.

This can be seen in greater detail in FIG. 2.

In the closed configuration **128**, the one-piece spout device **100b** may remain attached to a container opening **102** for safe storage of volatile fluids, the air-vent portion **116** portion allowing venting of fumes from a container.

The tether element **122** is preferably flexible and relaxed, the flexibility of the tether element allowing the tether element **122** to remain curved in the closed configuration **128**, which keeps the air-vent portion **116** engaged with the top opening **112** and is therefore unable to spring off of the top opening **112**.

The air-vent portion **116** preferably, but need not necessarily, includes a thumb tab **130** for easily pulling the air-vent portion **116** off of the top openings top opening **112**.

FIG. 2 illustrates a close-up view of a top end **108** of a one-piece spout device **200** in a closed configuration **128**,

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which can be used for safe storage of a container containing volatile fluids, such as gasoline.

Preferably, the spout body **104** comprises an outer surface **204** and an inner surface **206**.

Preferably, the top end **108** comprises one or more fin elements, said one or more fin elements encircling an outer surface of said one-piece lid, the fin elements being adapted to create a tight seal with said second end **120** of said air-vent portion **116**.

Preferably, the fin elements **202** are also made from resilient flexible material **132**.

The fin elements **202** create a tight seal with said air-vent portion **116** by pressing tightly against the second end **120** of the air-vent portion **116**, inhibiting it from leaking fluid from between the air-vent portion **116** and the top opening **112** of the spout body **104**.

The spout body **104** may further comprise rib elements **208** along the outer surface **204** of the spout body **104**, the rib elements **208** being sized and configured to fit into a vessel; for example, a gasoline tank of a car or a lawnmower.

FIG. **3** illustrates an embodiment of an air-vent portion **116**.

In an embodiment, the tether element **122** is a thin tether element **302**.

The thin tether element **302** is preferably flexible and relaxed, the flexibility enabling it to curve while in the closed configuration **128**.

The vent opening **124** preferably comprises a rib support element **304**, which provides support to the vent opening **124** when it is pushed into the top opening **112** of the one-piece spout device **200**.

Preferably, the air-vent portion **116** comprises a resilient flexible material **132**.

The resilient flexible material **132** preferably allows the air-vent portion **116** to fit tightly into the top opening **112** of the one-piece spout devices **100a** without leakage.

FIG. **4** illustrates a close-up view of a bottom end **106** of a one-piece spout device **400**, wherein the cap portion **114**, is releasably attached to a container **406** by twisting and/or screwing the cap portion **114** around a container opening **102**.

Containers **406** include, but are not limited to, fluid tanks, gas tanks, water tanks, and the like.

The one-piece spout device **400** is preferably compatible with a variety of different container and tank sizes.

For example, the one-piece spout device **400** can be releasably attached to standard 2, 5, and 10 gallon gas cans.

Preferably, the cap portion **114** comprises thread elements **402** at an upper end **408** and at a lower end **410** for tightening the cap portion **114** around the container threads **404** of the container opening **102**.

The thread elements **402** cooperate with the container threads **404** to form a self-sealing connection between the container **406** and the one-piece spout device **400**.

Given the flexibility of the one-piece spout device, there is no need for a gasket or o-ring to seal the cap portion **114** around the container opening **102**.

A mechanism for sealing the cap portion **114** around the container opening **102** comprises screwing or twisting the cap portion **114** around the container opening **102** such that the thread elements **402** cooperate with the container threads **404**.

Preferably, a sealing mechanism further comprises thread elements **402** at the upper end **408** jutting out further than the thread elements **402** at the lower end **410**, enabling the seal between the cap portion **114** and the container opening **102** to be tighter at the top, with a maximum interface between

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the container opening **102** and the cap portion **114** to occur at a lower interface **412** at the lower end **410** and a minimum interface at an upper interface **414** at the upper end **408**.

For example, in one embodiment, the maximum interface between the container opening **102** and the cap portion **114** is 0.02 inches.

Preferably, a sealing mechanism creates a tight seal between the cap portion **114** and the container opening **102**, which keeps fluid from leaking between the container and the one-piece spout device **400**.

FIG. **5** illustrates a side view of the cap portion **114**.

Preferably, the cap portion **114** is of a solid construction, made from resilient flexible material **132**, such that the cap portion **114** creates a tight seal around a container opening **102**.

The cap portion **114** preferably comprises a cap outer surface **504** and a cap inner surface **502**, the cap inner surface **502** comprising thread elements **402** which encircle the cap inner surface **502**. The thread elements **402** preferably cooperate with container threads **404** for twisting or screwing the cap portion **114** around a container opening **102**.

The following presents a simplified summary of the general inventive concepts described herein to provide a basic understanding of some aspects of the disclosure. This summary is not an extensive overview of the disclosure. It is not intended to restrict key or critical elements of embodiments of the disclosure or to delineate their scope beyond that which is explicitly or implicitly described by the following description and claims.

According to an embodiment of the invention, the container **406** may preferably comprise a fluid tank.

According to an embodiment of the invention, the fluid tank may preferably comprise a gas or water tank.

According to an embodiment of the invention, the spout body may preferably be conical in shape.

According to an embodiment of the invention, the spout body may preferably comprise an outer surface and an inner surface.

According to an embodiment of the invention, the one-piece spout may preferably comprise a resilient flexible material, the resilient flexible material being flexible over a wide range of temperatures.

According to an embodiment of the invention, the resilient flexible material may preferably, but need not necessarily, comprise nitrile rubber.

According to an embodiment of the invention, the cap portion may comprise thread elements for tightening the cap portion around the container opening, the thread elements cooperating with container threads on the container opening to create a tight seal between said cap portion and said container opening.

According to an embodiment of the invention, the top end may preferably comprise one or more fin elements, said one or more fin elements encircling an outer surface of said one-piece lid, the fin elements being adapted to create a tight seal with said second end of said air-vent portion.

According to an embodiment of the invention, the spout body may preferably be configured for insertion into a vessel.

According to an embodiment of the invention, the top opening may preferably be configured for insertion into a vessel.

According to an embodiment of the invention, the vessel may preferably, but need not necessarily, be a gasoline engine.

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According to an embodiment of the invention, the resilient flexible material may preferably, but need not necessarily, comprise nitrile rubber.

As one of ordinary skill in the art will readily appreciate from the disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

What is claimed is:

1. A one-piece spout device, attachable to a container opening, the one-piece spout comprising:
 - a spout body having a bottom end with a bottom opening and a top end, the top end having an outer surface and a top opening;
 - a cap portion affixed to the bottom end, the cap portion being sized and adapted to releasably connect said one-piece spout to said container;
 - an air-vent portion comprising a first end and a second end, the first end being affixed to the top end of said spout body, and the second end being attachable to the top opening of said top end, the second end comprising a vent opening for partially covering said top opening; wherein the first end and the second end of said air-vent portion are attached by a tether element;
 - wherein the top end of the spout body comprises one or more fin elements, said one or more fin elements

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encircling the outer surface, the fin elements being adapted to create a tight seal with said second end of said air-vent portion; and

wherein the cap portion and the spout body are of a one-piece molded construction formed from nitrile rubber.

2. The one-piece spout of claim 1, wherein the container comprises a fluid tank.

3. The one-piece spout of claim 2, wherein the fluid tank is a gas tank.

4. The one-piece spout of claim 2, wherein the fluid tank is a water tank.

5. The one-piece spout of claim 1, wherein the spout body is conical in shape.

6. The one-piece spout of claim 1, wherein the spout body comprises an outer surface and an inner surface.

7. The one-piece spout of claim 1, wherein the nitrile rubber is flexible over a wide range of temperatures.

8. The one-piece spout of claim 1, wherein the cap portion comprises thread elements for tightening the cap portion around the container opening, the thread elements cooperating with container threads on the container opening to create a tight seal between said cap portion and said container opening.

9. The one-piece spout of claim 1, the spout body being configured for insertion into a vessel.

10. The one-piece spout of claim 9, the top opening being configured for insertion into a vessel.

11. The one-piece spout of claim 9, the vessel being a gasoline engine.

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